

# E-Commerce Sales Insights System

SQL-first relational analytics + local stakeholder dashboard

|                 |                                                                     |
|-----------------|---------------------------------------------------------------------|
| PORTFOLIO STACK | SQLite • SQL • Python Standard Library                              |
| DATA PERIOD     | January - June 2026                                                 |
| DEPLOYMENT      | Local, serverless database; no Oracle/MySQL/PostgreSQL installation |
| PRIMARY USE     | Sales, marketing, customer, product and operations analytics        |

## Project Summary

This project models a complete e-commerce analytics environment: customers, product catalog, orders, order items, payments, marketing channels, spend, website sessions, shipping, returns, and inventory. SQL views and analytical queries convert transaction-level data into decision-ready KPIs, while a lightweight local dashboard presents the results for stakeholders.

**Why SQLite?** The original concept is suitable for PostgreSQL, but this packaged version uses SQLite so the database ships as a single file and runs without a database server. The relational model, SQL joins, constraints, views, indexes, aggregations and CTE-based analytics remain directly relevant to SQL/BI interviews.

## Executive KPI Snapshot

| Paid Orders | Net Revenue | AOV     | Units Sold | Gross Margin | Repeat Rate |
|-------------|-------------|---------|------------|--------------|-------------|
| 838         | \$53,343.76 | \$63.66 | 1,786      | 64.58%       | 39.34%      |

# 1. Architecture & Design Decisions

|                       |                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------|
| 1. Data Layer         | SQLite relational database (ecsales.db)                                                      |
| 2. Modeling Layer     | Normalized tables with PK/FK/CHECK constraints                                               |
| 3. Semantic Layer     | Reusable SQL views: paid order lines, order summary, channel daily, customer LTV             |
| 4. Analytics Layer    | 15 business queries for sales, marketing, cohorts, refunds, shipping, payments and inventory |
| 5. Presentation Layer | Local Python dashboard with Overview, Marketing, Products, Customers, Operations and SQL Lab |

## Database entities

The model contains 12 core entities/views supporting both commercial and operational analysis. Orders are separated from order items so one order can contain multiple products; product cost is kept in Products while transaction selling price and discount are stored at line level. Marketing spend and website sessions are daily facts by channel, enabling ROAS, CAC and conversion analysis.

| Entity           | Purpose                                                     | Rows  |
|------------------|-------------------------------------------------------------|-------|
| customers        | Customer profile, geography, signup and acquisition channel | 700   |
| products         | SKU, category, unit cost and list price                     | 31    |
| orders           | Order header, status, channel and destination               | 949   |
| order_items      | Line-level quantity, price and discount                     | 1,576 |
| payments         | Payment amount, method and status                           | 949   |
| shipping         | Courier, promise, delivery date/cost/status                 | 949   |
| returns          | Return reason and refund value                              | 47    |
| website_sessions | Sessions, carts, checkouts and conversions by day/channel   | 1,086 |
| channel_spend    | Daily channel spend, impressions and clicks                 | 1,086 |

## 2. KPI & Business Logic

| Metric             | Definition                                                                | Business use                                                    |
|--------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------|
| Net Revenue        | Sum of line revenue for PAID orders: (unit price × quantity) × line count | Totals realized sales excluding cancelled/refunded order status |
| AOV                | Net revenue / paid order count                                            | Measures typical basket value                                   |
| Gross Profit       | Line revenue - product cost                                               | Measures product economics before marketing/shipping overhead   |
| ROAS               | Attributed revenue / channel spend                                        | Compares marketing revenue efficiency                           |
| CAC                | Channel spend / new customers                                             | Evaluates acquisition cost                                      |
| Conversion Rate    | Conversions / website sessions                                            | Measures funnel efficiency                                      |
| Repeat Rate        | Customers with 2+ paid orders / purchasing customers                      | Signals retention and customer quality                          |
| 30-day Cohort Rate | Customers who reorder within 30 days of first purchase                    | Evaluates repeat behavior                                       |
| Late Delivery Rate | Delivered shipments after promised date / delivered shipments             | Measures fulfillment service level                              |

### Example analytical SQL

```
SELECT channel_name,  
       ROUND(SUM(revenue),2) AS revenue,  
       ROUND(SUM(spend),2) AS spend,  
       ROUND(SUM(revenue)/NULLIF(SUM(spend),0),2) AS roas  
FROM v_channel_daily  
GROUP BY channel_name  
ORDER BY roas DESC;
```

## 3. Marketing Performance

| Channel        | Revenue     | Spend       | ROAS | Conversion % |
|----------------|-------------|-------------|------|--------------|
| Organic Search | \$7,055.14  | \$6,821.77  | 1.03 | 3.72%        |
| Email          | \$6,816.51  | \$8,617.44  | 0.79 | 4.88%        |
| Meta Ads       | \$13,429.51 | \$67,509.78 | 0.20 | 2.91%        |
| Google Ads     | \$14,085.75 | \$81,806.24 | 0.17 | 3.36%        |
| Influencer     | \$2,883.45  | \$34,374.14 | 0.08 | 2.62%        |
| Direct         | \$9,073.40  | \$0.00      | -    | 4.13%        |

Interpretation: channels should not be ranked on revenue alone. Revenue, spend, ROAS, CAC and conversion rate answer different questions about scale, efficiency and funnel quality. Organic/owned channels can show undefined or unusually high ROAS when direct media spend is zero or near zero, so the dashboard preserves NULL semantics rather than inventing a value.

## 4. Product, Customer & Operations Insights

### Top products by paid revenue

| Product             | Category    | Units | Revenue    |
|---------------------|-------------|-------|------------|
| Mechanical Keyboard | Electronics | 63    | \$5,291.12 |
| Smart Watch         | Electronics | 60    | \$4,537.84 |
| Denim Jacket        | Fashion     | 60    | \$2,964.32 |
| Portable Speaker    | Electronics | 46    | \$2,783.52 |
| Power Bank 20000mAh | Electronics | 68    | \$2,522.72 |

### Customer analytics

The database contains 516 purchasing customers, of whom 203 made at least two paid purchases. The repeat customer rate is 39.34%. The customer LTV view also exposes paid order count, lifetime revenue, average order value, first order date and last order date for ranking high-value customers.

### Shipping performance

| Courier      | Delivered | Avg days | Late % |
|--------------|-----------|----------|--------|
| DTDC         | 210       | 4.73     | 43.81% |
| Ecom Express | 189       | 4.75     | 43.92% |
| BlueDart     | 204       | 4.70     | 44.12% |
| Delhivery    | 235       | 4.73     | 46.81% |

## 5. Dashboard & User Experience

| Page       | Decision support                                                              |
|------------|-------------------------------------------------------------------------------|
| Overview   | Executive revenue, AOV, margin, product and regional performance with filters |
| Marketing  | Channel revenue, spend, ROAS, CAC, sessions and conversion                    |
| Products   | Category/product profitability and reorder-risk inventory                     |
| Customers  | Lifetime value, repeat purchase and 30-day cohort retention                   |
| Operations | Courier service levels, returns/refunds and payment mix                       |
| SQL Lab    | Read-only SELECT/WITH queries for demonstrating SQL exploration live          |

**Run locally:** extract the ZIP and double-click `run_dashboard.bat`. The application uses Python standard-library HTTP services and the bundled SQLite database; no third-party Python package or SQL server is required.

## 6. SQL Engineering & Optimization

The project uses indexes on frequent join/filter keys, including order status/date, customer ID, channel ID, order-item order/product IDs, product category, marketing spend date/channel, shipping order and return product. Views reduce duplication of business logic and create a stable semantic layer for the dashboard.

### Key SQL techniques demonstrated

- Primary keys, foreign keys and CHECK constraints
- INNER JOIN and LEFT JOIN across transactional/reference tables
- SUM, COUNT, AVG and conditional aggregation
- CTEs for cohort and multi-stage calculations
- Subqueries and reusable views
- Date grouping and date-difference logic
- NULL-safe division using NULLIF
- Indexes for join/filter performance
- Read-only ad hoc querying in the SQL Lab

### Portability note

SQLite uses `date()`, `strftime()` and `julianday()` for this package. A PostgreSQL migration would replace these with date casts, `DATE_TRUNC` and interval/date arithmetic. The relational design and KPI equations do not change.

## 7. Interview Positioning

### Recommended one-line description

**“I built a SQL-first E-Commerce Sales Insights System with a normalized relational database, reusable analytics views, marketing and cohort KPIs, operational metrics, and a lightweight stakeholder dashboard.”**

### What to emphasize

Explain the business question before the query. For example: “To compare acquisition channels, I combined session, spend, new-customer and attributed sales facts, then calculated conversion, ROAS and CAC.” Also be explicit that SQLite was chosen for portable demonstration; do not describe it as a cloud data warehouse.

### Next production upgrades

PostgreSQL or a cloud warehouse; automated ETL; Power BI/Tableau semantic models; scheduled refresh; role-based access; data-quality checks; multi-touch attribution; segmentation, forecasting and recommendation features.

## Appendix A - Deliverables

| File/Folder                  | Purpose                                          |
|------------------------------|--------------------------------------------------|
| database/ecsales.db          | Ready-to-use relational database                 |
| sql/01_schema.sql            | DDL for core entities and constraints            |
| sql/02_views_indexes.sql     | Semantic views and optimization indexes          |
| sql/03_analytics_queries.sql | 15 interview/business analytics queries          |
| data/*.csv                   | Source-table exports for inspection or BI import |
| app.py                       | Local dashboard application                      |
| run_dashboard.bat            | Windows one-click launcher                       |
| docs/Interview_QA.md         | 10 interview-ready questions and answers         |
| docs/Project_Report.pdf      | This project report                              |

### Dataset disclaimer

All data in this portfolio package is synthetic and generated for demonstration. Example names, emails, orders, marketing activity, shipping events and return records do not represent real customers or companies.